

Physio. -CARDIOTHORACIC MEDICIN and SURGERY

DR. Azza Hamed AbdElwahab

1- About the pulmonary vascular resistance, Which of the following is incorrect?

- a. It is decreased during exercise.
- b. May lead to pulmonary hypertension.
- c. It is increased in lung diseases.
- d. **It is decreased in hypoxia.**

2- If the cardiac output is doubled, what happens to pulmonary arterial pressure?

- a. Becomes 2 times normal.
- b. **Kept constant.**
- c. Becomes 4 times normal.
- d. Markedly decreased.

3- Why the pulmonary artery BP is lower than the systemic arterial BP?

- a) The output of the right ventricle is less than that of the left ventricle.
- b) **The pulmonary resistance is less than the systemic resistance.**
- c) The left ventricular wall is thicker than that of the right ventricle.
- d) The aorta is more elastic than the pulmonary artery.

4- Which of the following is true regarding pulmonary circulation?

- A. It is the larger of the two circulatory systems.
- B. **It is a low-pressure system that allows for improved gas exchange.**
- C. The system functions with an increased arterial pressure to circulate through the distal parts of the body.
- D. It consists of the left side of the heart, the aorta, and its branches.

5- A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her left ventricular volume unchanged?

- A. Rapid filling
- B. **Isovolumetric contraction**
- C. Ventricular ejection

D. Atrial systole

6- Why the S A node is the normal pace maker of the heart?

- (a) Is the most rapidly discharging part.
- (b) Is the most richly supplied by nerve endings.
- (c) Located in the atrium.
- (d) Is the most rapidly conducting part.

7- Which of the followings characterizes the S-A node?

- (a) Cells within the S-A node act as heart pace maker because their membrane depolarized to threshold and initiate an action potential.
- (b) Acetyl choline increases the slope of the pace-maker potential.
- (c) Sympathetic stimulation decreases the slope of the pace maker potential.
- (d) The pace-maker cells within the S A node are neurons rather than myocytes.

? What is the cardiac tissue having slowest rhythmicity-8

- .SAN without vagal tone (a)
- . SAN with vagal tone (b)
- (c) Ventricular muscle.
- (d) AVN.

9- In which phase of cardiac cycle, the second heart sound can be heard?

- a) Atrial contraction phase.
- b) Isometric relaxation phase.
- c) Maximum ejection phase.
- d) Reduced filling phase.

10- Which of the following events occur during maximum ejection phase of cardiac cycle?

- a) The first heart sound is heard.
- b) The T-wave of the ECG is at its peak.
- c) The ventricular volume is constant.
- d) The ventricular pressure reaches maximum.

- 1. Enumerate factors affecting pulmonary blood pressure and flow**
- 2. What is the meaning of Ventilation/Perfusion Ratio:**
- 3. Determine the ionic basis of ventricular action potential**
- 4- What is the cardiac ejection fraction?**
- 5. Describe relationship between heart rate (HR), cardiac filling, and cardiac output (CO)**
- 6. What do you know about lung zones?**